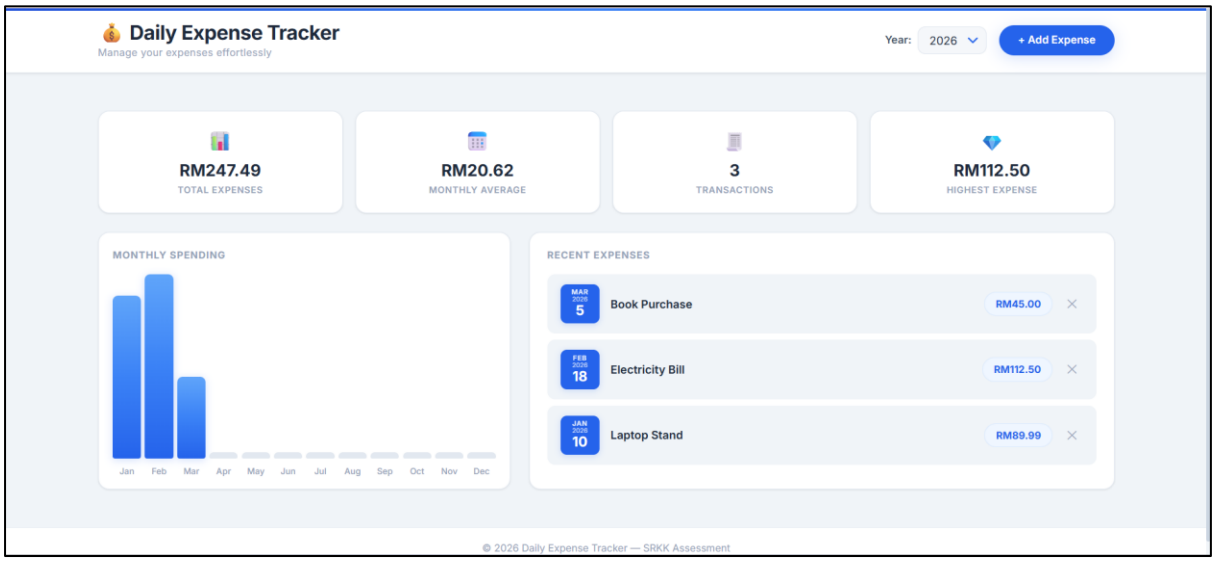
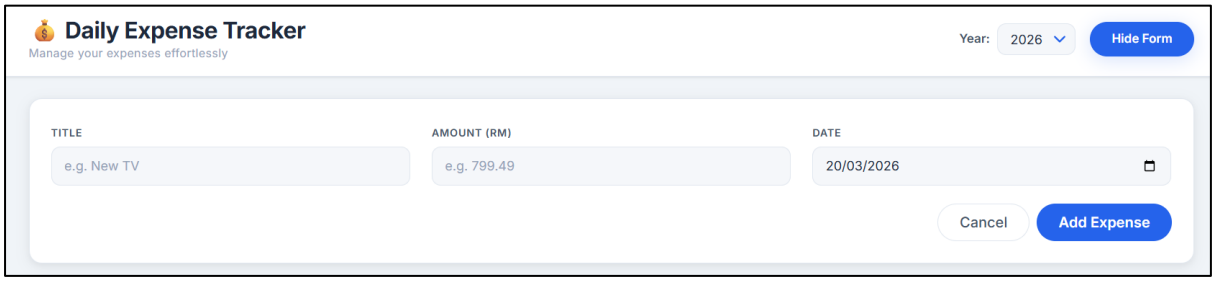
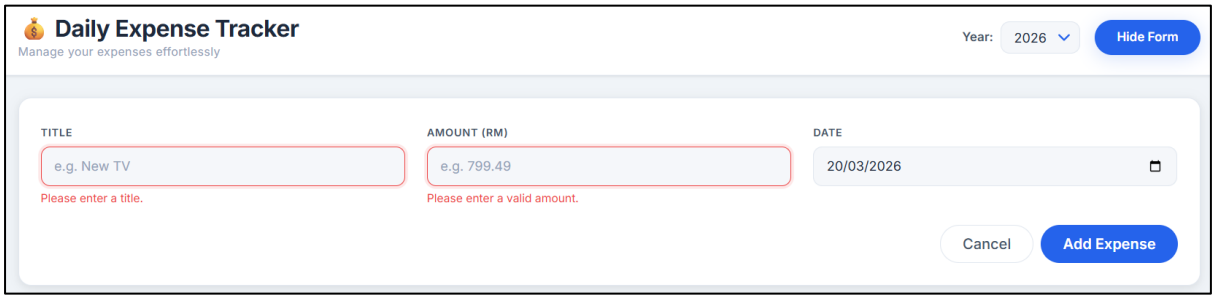
INTERFACE DESIGN:



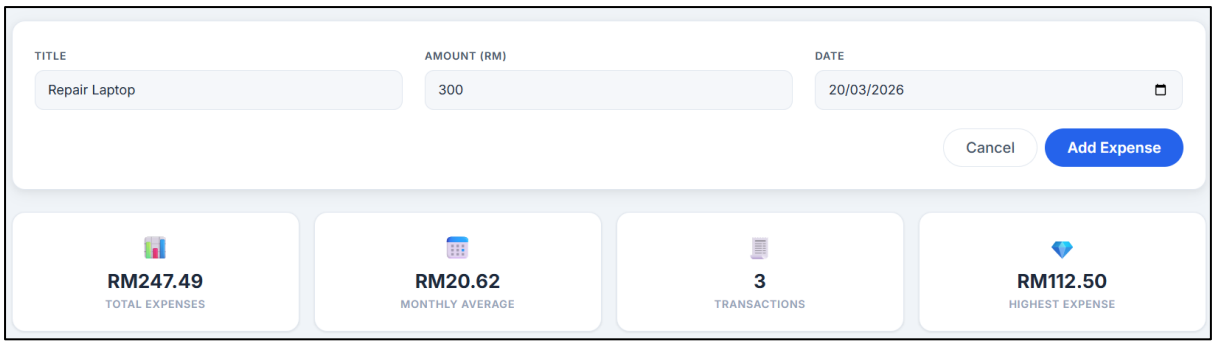
Main Dashboard View: This is the primary landing page. It displays a modern, card-based dashboard with four **Key Performance Indicators (KPIs)**: Total Expenses, Monthly Average, total Number of Transactions, and the Highest Expense. It also features a "Monthly Spending" bar chart for visual data analysis.



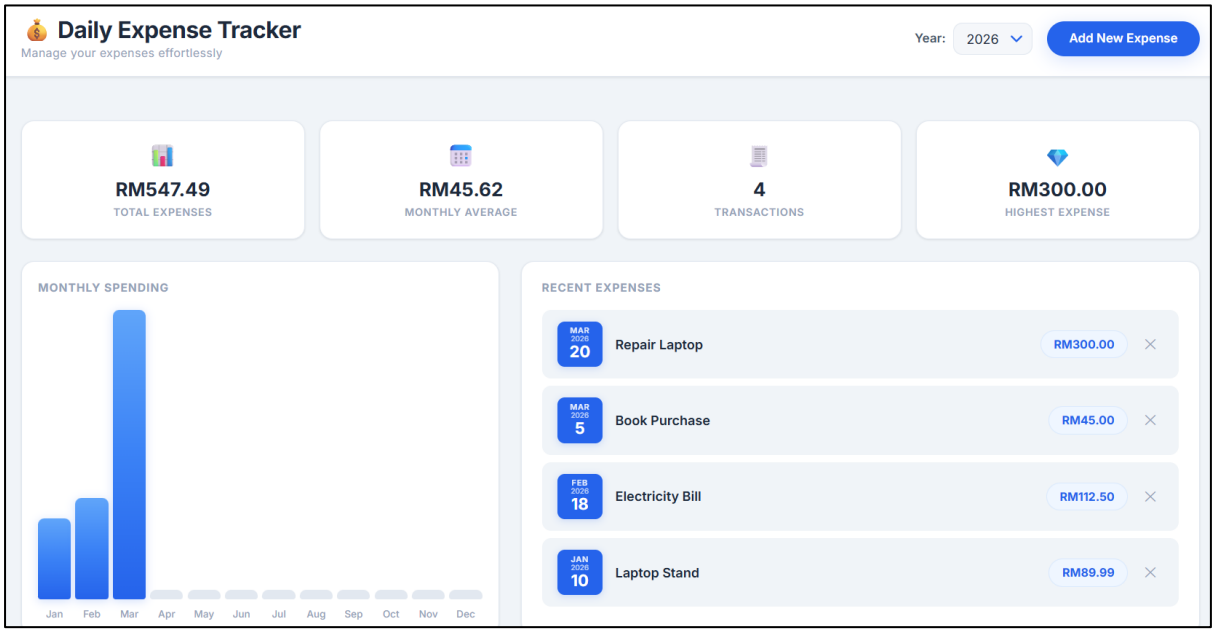
Add Expense Form: This shows the responsive form section. When the user clicks the "Add Expense" button, the form expands. I used **Bootstrap** to ensure this layout is clean and adapts to different screen sizes.



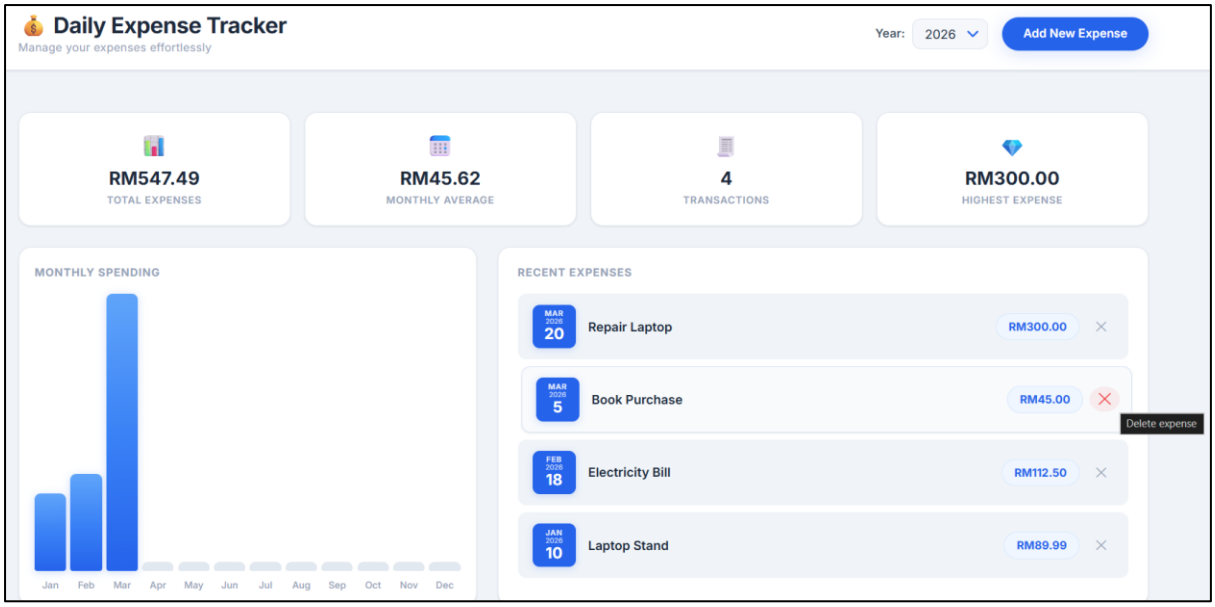
Input Validation: This image demonstrates the **real-time validation logic**. If a user attempts to submit without entering a title or a valid amount, the system provides immediate visual feedback (red borders and error text), preventing "empty" or "null" data from entering the system.



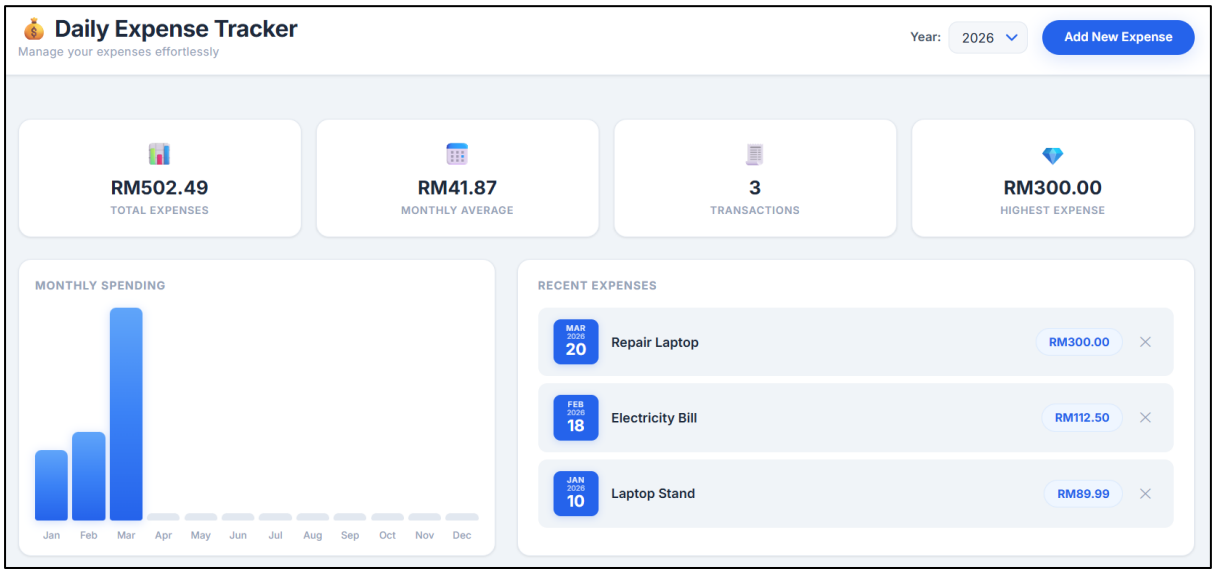
Data Entry: Shows the user filling out a specific expense (Repair Laptop - RM300). This illustrates the transition before the data is processed by the JavaScript logic.



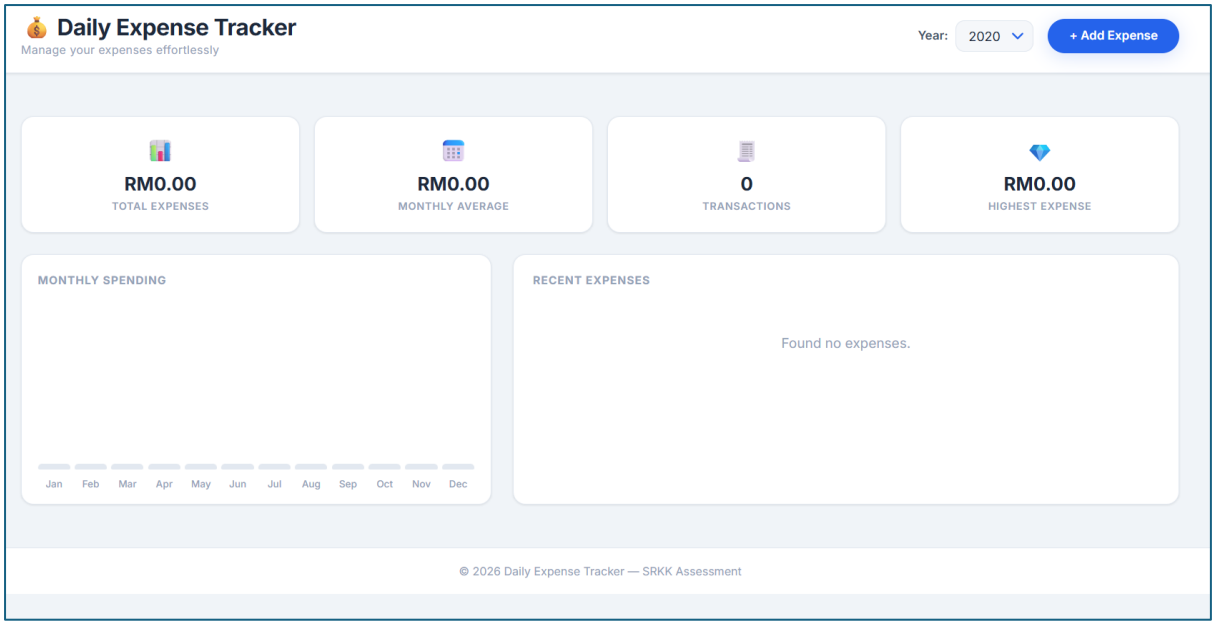
Successful Data Integration: This is the result after clicking "Add Expense." The system performs an **asynchronous update**: The "Total Expenses" card jumps from RM247.49 to **RM547.49**. The "Transactions" count increases to **4**. The **Bar Chart for March** automatically grows to reflect the new total. This demonstrates advanced state management in JavaScript.



The Delete Feature: This highlights one of the "additional" creative features is added. By hovering over an expense, a "Delete expense" tooltip appears, showing a user-friendly way to manage the data.



Dynamic Recalculation: This image shows the dashboard after an expense ("Book Purchase") has been deleted. Note how all the cards and the bar chart **automatically recalculated** in real-time. This ensures the user always sees the most accurate financial data without needing to refresh the page.



This image demonstrates the system's **robust filtering logic** and **empty state handling**, specifically fulfilling the requirement to show a fallback message when no data is available for a selected period. This system implemented a dynamic filtering system that handles empty states gracefully. As shown, selecting a year with no data (2020) triggers a 'Found no expenses' placeholder and resets all analytical charts and KPI metrics to zero in real-time.

These screenshots demonstrate a full **CRUD (Create, Read, Delete)** lifecycle. I prioritized **data integrity** through robust validation and provided a high-end **User Experience (UX)** by ensuring all visual elements including the charts and KPI cards update dynamically as the data changes.

Source Code: https://github.com/Shahrulzaman/Daily-Expense-Tracker.git